John Nacion

johnnacion.dev
github.com/nuttycream
me@johnnacion.dev

Summary

USAF veteran and now Test Engineer with a B.S. in Computer Science: automated test and Rust/C tooling on one side, cleanroom optics integration and non-conformance traceability on the other.

Skills

Languages & Software:	Rust, C/C++, C#, Python, Git, GNU/Linux, NixOS, Vim/Neovim, embedded GPIO/I2C/SPI, DMA
Test Equipment:	Oscilloscopes, DMM, accelerometers, electron and optical microscopes, electronic leak detectors, pressure gauges, calipers, torque and precision hand tools
Manufacturing & Quality:	ISO 9001, Class 100/10K cleanroom, ESD, SOP/work-instruction compliance, non-conformance (NC) logging, test traceability, root-cause analysis

Experience

Test Engineer II (Broadband Plasma Division, Illumination Group) - KLA **May 2026 - Present**

- Execute integration and acceptance testing on broadband-plasma wafer-inspection tools, running automated test scripts and analyzing output against engineering specs to disposition optical and electro-mechanical sub-assemblies.
- Isolate defects that span software, optics, and hardware boundaries; characterize failures with oscilloscopes, accelerometers, and leak detectors, then drive rework and re-test until the assembly passes spec.
- Log, triage, and escalate non-conformances through KLA's internal test/quality software; work directly with Mechanical Design Engineers to close root causes and make ship dates for fabs including TSMC, Samsung, SK Hynix, and Micron.
- Capture test data, defects, and rework with full traceability, and surface recurring failure patterns to engineering for process improvement.
- Assemble and align precision optical and mechanical components to arcsecond angular and micrometer-to-nanometer positional tolerances in an ISO 9001, Class 100/10K cleanroom under ESD controls.

R&D Operator II - The Bot Company **Feb 2026 - May 2026**

- Operated robotic platforms across multiple field sites, executing structured data-collection protocols to capture ground-truth training data for early-stage AI models.
- Ranked among the top operators on every daily metric (data volume, accuracy, throughput) in a performance-driven environment.

Robotics Operator - armstrong.ai **Jun 2025 - Jan 2026**

- Operated multiple robots asynchronously through a proprietary dashboard and command-line tools for system-level control.
- Diagnosed and resolved day-to-day operational issues, coordinating with on-site technicians in real time to maintain continuous 24/7 uptime.

Security Forces - United States Air Force **Jun 2015 - Jun 2018**

- Entry Controller for the second-largest non-nuclear joint installation in the Department of Defense; maintained security of the local SCIF and performed route clearance for visiting senior officials.
- Responded to and investigated 100+ unannounced alarm activations, incidents, criminal offenses, and traffic incidents.

Projects

Rust GPIO Controller Web-Application **GitHub**

- Maintain a GPIO controller web-app with a Rust backend on an async Tokio runtime. GPIO control is implemented through direct memory-mapped register access on the gpiomem device; built for Raspberry Pi but accepts any GPIO device with a valid base address.

Omnidirectional Robot with Obstacle Avoidance/Tracking **Article**

- Built a three-wheeled robot that autonomously follows multi-colored lines using omnidirectional movement. Written in C on a custom DMA library supporting GPIO, I2C, and SPI; ultrasonic sensors detect, track, and avoid obstacles along a dynamically computed path.

Cross-platform GUI to Modify Game Saves **GitHub**

- Developed an open-source save editor for a popular PC game with 50k+ downloads across modding forums and GitHub. C# with Avalonia; parses and modifies XML via System.Xml with XPath and dotNET's task-based async library.

Version Control Automation using Large Language Models **GitHub**

- Built a CLI that drives local and hosted LLM providers to generate commit messages from diffs using commit message conventions, plan rebases against a target branch, amend commits, and find commits via natural-language queries.

Education

Bachelor of Science, Computer Science — San Francisco State University, 2025